

Pre Mid Sem Examination

Sep 20th, 2024

Time: 1 hour 15 minutes

Max Marks: 50

Instructions:

1. Attempt all the five questions. All questions carry equal marks (parts may have different marks).
2. All your intermediate steps and calculations must be clearly shown.
3. Marks for proof-type questions will depend on the logical progression of the steps. You may quote without proof any proposition or theorem covered in the lectures and tutorials but it must be clearly identified. You are not allowed to use determinant. Any other results used must be proved.
4. Write clearly and identify each part of your answers for the benefit of graders.

Problem 1. 1. Find if the system of equations $A\bar{x} = \bar{b}$ is consistent (show reasons), where

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix} \quad \bar{b} = \begin{bmatrix} 4 \\ 3 \\ 1 \\ 2 \end{bmatrix}.$$

$$\begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix} \quad 142 - 110 - 38 + 16$$

2. If the answer is yes, find the solution(s).
3. How many solutions do you have?

Problem 2. 1. Find an LU factorization of $A = \begin{bmatrix} 3 & 1 & 3 & -4 \\ 6 & 4 & 8 & -10 \\ 3 & 2 & 5 & -1 \\ -9 & 5 & -2 & -4 \end{bmatrix}$

$$\begin{bmatrix} 19 \\ 3 \\ -22 \\ 19 \\ -4 \end{bmatrix}$$

2. Use the LU factorization method to solve the linear system of equations $A\bar{x} = \bar{b}$ where A is given in previous part and $\bar{b}$ is the vector given below:

$$\bar{b} = \begin{bmatrix} 2 \\ 6 \\ 6 \\ 3 \end{bmatrix}$$

Problem 3. In each of the following parts, find whether the set W is a subspace of the vector space V . (Prove or disprove)

1. $V = \mathbb{R}^2$, W is the graph of the function $f(x) = x + 3x^2$.

Recall the graph of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined as the set of points $\{(x, y) : y = f(x)\}$.

2. $V = C[0, 1]$, the space of all real valued continuous functions defined on $[0, 1]$ and W is the set of all real valued continuous functions f defined on $[0, 1]$ such that $f(\frac{1}{2}) \in \mathbb{Q}$. ($\mathbb{Q}$ is the set of rational numbers.)

3. $V = \mathbb{R}^\infty$, the set of all real sequences and $W = C_0$, i.e., the set of all real sequences converging to zero.

4. $V = \mathbb{R}^n$, $W = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : x_1 + x_2 + \dots + x_n = 0\}$

Problem 4. 1. If A is a 4×4 matrix such that $5A^4 - 3A^3 + 8A^2 - 7A + 29I = 0$, find if A is invertible or not. If A is invertible write A^{-1} in terms of A .

2. Does $\mathbb{Z}_{21}$ contain any divisor of zero? Give reasons. Hence conclude if $\mathbb{Z}_{21}$ is a field or not.

Problem 5. 1. If a system of equations $A\bar{x} = \bar{b}$ is changed into a new system $C\bar{x} = \bar{d}$ by a row replacement operation, show that the two systems have the same set of solutions.

2. If A and B are $m \times n$ matrices over $\mathbb{R}$ and $A\bar{x} = B\bar{x}$ for all $\bar{x} \in \mathbb{R}^n$, show that $A = B$.